

# Cell-Type-Specific Biomarkers in Human Liver Fibrosis

Reproducible single-cell discovery pipeline, full report

2026-06-07

## Overview

This report is generated by the liver-fibrosis pipeline. It summarizes QC, integration, annotation, donor-aware differential expression, pathway and cell-cell communication mechanism analysis, and the AI/ML biomarker prioritization that yields the ranked candidate table. Method rationale and citations are in `docs/methods.md`; written answers to the screening questions are in `docs/written_answers.md`.

## QC summary

### gse136103

|    | sample     | n_cells_before | n_cells_after | pct_kept | doublets_removed | mito_pct_ceiling |
|----|------------|----------------|---------------|----------|------------------|------------------|
| 0  | GSM4041150 | 1863           | 1478          | 79.3     | 4                | 8.00             |
| 1  | GSM4041151 | 1039           | 655           | 63.0     | 0                | 8.00             |
| 2  | GSM4041152 | 478            | 299           | 62.6     | 1                | 8.00             |
| 3  | GSM4041153 | 6571           | 4762          | 72.5     | 1056             | 8.00             |
| 4  | GSM4041154 | 4688           | 4128          | 88.1     | 1                | 8.00             |
| 5  | GSM4041155 | 3295           | 2429          | 73.7     | 39               | 5.87             |
| 6  | GSM4041156 | 3230           | 2550          | 78.9     | 4                | 7.06             |
| 7  | GSM4041157 | 1156           | 820           | 70.9     | 5                | 8.00             |
| 8  | GSM4041158 | 4937           | 4123          | 83.5     | 1                | 8.00             |
| 9  | GSM4041159 | 3306           | 2826          | 85.5     | 1                | 6.37             |
| 10 | GSM4041160 | 5122           | 4044          | 79.0     | 2                | 5.28             |
| 11 | GSM4041161 | 2380           | 1764          | 74.1     | 9                | 6.35             |
| 12 | GSM4041162 | 1686           | 1231          | 73.0     | 1                | 8.00             |
| 13 | GSM4041163 | 2046           | 1231          | 60.2     | 3                | 8.00             |
| 14 | GSM4041164 | 3412           | 2698          | 79.1     | 4                | 8.00             |
| 15 | GSM4041165 | 3673           | 3031          | 82.5     | 2                | 6.19             |
| 16 | GSM4041166 | 1985           | 1568          | 79.0     | 1                | 8.00             |
| 17 | GSM4041167 | 3602           | 2955          | 82.0     | 0                | 7.64             |
| 18 | GSM4041168 | 4819           | 3801          | 78.9     | 0                | 3.96             |
| 19 | GSM4041169 | 2922           | 2061          | 70.5     | 5                | 5.17             |

### gse244832

|   | sample | n_cells_before | n_cells_after | pct_kept | doublets_removed | mito_pct_ceiling |
|---|--------|----------------|---------------|----------|------------------|------------------|
| 0 | JB288  | 7492           | 5650          | 75.4     | 87               | 0.60             |
| 1 | JB289  | 3447           | 2152          | 62.4     | 12               | 0.11             |
| 2 | JB290  | 9148           | 6693          | 73.2     | 10               | 0.57             |
| 3 | JB303  | 8739           | 6764          | 77.4     | 28               | 0.32             |
| 4 | JB304  | 6531           | 3702          | 56.7     | 35               | 0.00             |
| 5 | JB305  | 7134           | 4408          | 61.8     | 19               | 0.10             |
| 6 | JB317  | 6223           | 4350          | 69.9     | 5                | 0.06             |

|    | sample | n_cells_before | n_cells_after | pct_kept | doublets_removed | mito_pct_ceiling |
|----|--------|----------------|---------------|----------|------------------|------------------|
| 7  | JB318  | 6867           | 5158          | 75.1     | 2                | 0.43             |
| 8  | JB319  | 7541           | 5447          | 72.2     | 4                | 0.18             |
| 9  | JB320  | 8056           | 5793          | 71.9     | 3                | 0.08             |
| 10 | JB321  | 7325           | 5432          | 74.2     | 0                | 0.17             |
| 11 | JB322  | 8703           | 6863          | 78.9     | 5                | 0.56             |
| 12 | JB336  | 4260           | 3360          | 78.9     | 23               | 0.28             |
| 13 | JB337  | 7960           | 6320          | 79.4     | 5                | 0.20             |
| 14 | JB338  | 7244           | 5412          | 74.7     | 2                | 0.17             |
| 15 | JB339  | 6936           | 5308          | 76.5     | 10               | 0.35             |
| 16 | JB340  | 7452           | 5835          | 78.3     | 6                | 0.34             |
| 17 | JB341  | 7165           | 5327          | 74.3     | 1                | 0.34             |

## Integration (scIB) and annotation

### gse136103

|   | n_cells | n_batches | batch_silhouette | condition_silhouette | condition_knn_accuracy | fibrosis_signal_guard_pass |
|---|---------|-----------|------------------|----------------------|------------------------|----------------------------|
| 0 | 48454   | 20        | -0.144           | 0.021                | 0.641                  | True                       |

|    | leiden | n_cells | cell_type        | compartment            | frac_fibrotic_F3plus |
|----|--------|---------|------------------|------------------------|----------------------|
| 0  | 0      | 7302    | T_NK             | lymphoid               | 0.499                |
| 1  | 1      | 6661    | T_NK             | lymphoid               | 0.287                |
| 2  | 2      | 6547    | T_NK             | lymphoid               | 0.301                |
| 3  | 3      | 5056    | Monocyte_MoMac   | macrophage_monocyte    | 0.502                |
| 4  | 4      | 4566    | T_NK             | lymphoid               | 0.347                |
| 5  | 5      | 3838    | T_NK             | lymphoid               | 0.354                |
| 6  | 6      | 3314    | Endothelial_vasc | endothelial            | 0.722                |
| 7  | 7      | 1969    | Endothelial_vasc | endothelial            | 0.642                |
| 8  | 8      | 1911    | HSC_mesenchyme   | stellate_myofibroblast | 0.282                |
| 9  | 9      | 1758    | Kupffer_resident | macrophage_monocyte    | 0.357                |
| 10 | 10     | 1462    | Cholangiocyte    | cholangiocyte          | 0.521                |
| 11 | 11     | 1350    | LSEC             | endothelial            | 0.250                |
| 12 | 12     | 1220    | B_plasma         | lymphoid               | 0.653                |
| 13 | 13     | 886     | T_NK             | lymphoid               | 0.221                |
| 14 | 14     | 237     | Endothelial_vasc | endothelial            | 0.835                |
| 15 | 15     | 229     | DC               | myeloid_other          | 0.655                |
| 16 | 16     | 148     | B_plasma         | lymphoid               | 0.486                |

### gse244832

|   | n_cells | n_batches | batch_silhouette | condition_silhouette | condition_knn_accuracy | fibrosis_signal_guard_pass |
|---|---------|-----------|------------------|----------------------|------------------------|----------------------------|
| 0 | 93974   | 18        | -0.176           | -0.053               | 0.546                  | False                      |

|    | leiden | n_cells | cell_type        | compartment            | frac_fibrotic_F3plus |
|----|--------|---------|------------------|------------------------|----------------------|
| 0  | 0      | 15961   | Hepatocyte       | hepatocyte             | 0.501                |
| 1  | 1      | 15528   | Hepatocyte       | hepatocyte             | 0.439                |
| 2  | 2      | 12761   | Hepatocyte       | hepatocyte             | 0.435                |
| 3  | 3      | 12270   | Hepatocyte       | hepatocyte             | 0.633                |
| 4  | 4      | 8483    | LSEC             | endothelial            | 0.583                |
| 5  | 5      | 7028    | Hepatocyte       | hepatocyte             | 0.441                |
| 6  | 6      | 6761    | LSEC             | endothelial            | 0.590                |
| 7  | 7      | 6492    | HSC_mesenchyme   | stellate_myofibroblast | 0.824                |
| 8  | 8      | 3364    | Kupffer_resident | macrophage_monocyte    | 0.598                |
| 9  | 9      | 1607    | Hepatocyte       | hepatocyte             | 0.127                |
| 10 | 10     | 1309    | HSC_mesenchyme   | stellate_myofibroblast | 0.906                |
| 11 | 11     | 861     | Hepatocyte       | hepatocyte             | 0.553                |

|    | leiden | n_cells | cell_type        | compartment   | frac_fibrotic_F3plus |
|----|--------|---------|------------------|---------------|----------------------|
| 12 | 12     | 754     | Cholangiocyte    | cholangiocyte | 0.893                |
| 13 | 13     | 263     | Endothelial_vasc | endothelial   | 0.992                |
| 14 | 14     | 219     | Hepatocyte       | hepatocyte    | 0.982                |
| 15 | 15     | 155     | Hepatocyte       | hepatocyte    | 0.129                |
| 16 | 16     | 98      | Hepatocyte       | hepatocyte    | 1.000                |
| 17 | 17     | 60      | Hepatocyte       | hepatocyte    | 1.000                |

## Ranked biomarker / target candidates

The headline list is gated on differential-expression significance (compartment-level DESeq2 *or* a disease-associated-subcluster marker) and on cell-type specificity, so each candidate is a genuine, compartment-specific disease signal rather than an abundant lineage marker. Candidates flagged `in_disease_niche` are recovered from a disease-associated subpopulation that compartment-level pseudobulk averages out (e.g. scar-associated endothelium / macrophages).

|     | rank | gene    | compartment            | composite | log2FoldChange | padj  | accessibility_class | druggability | in_disease_niche |
|-----|------|---------|------------------------|-----------|----------------|-------|---------------------|--------------|------------------|
| 0   | 1.0  | PDGFRA  | stellate_myofibroblast | 0.671     | 3.608          | 0.000 | surface             | 0.90         | True             |
| 1   | 2.0  | MMP7    | cholangiocyte          | 0.645     | 1.956          | 0.000 | secreted            | 0.10         | False            |
| 2   | 3.0  | SPP2    | cholangiocyte          | 0.644     | 4.315          | 0.001 | secreted            | 0.10         | False            |
| 3   | 4.0  | ACKR1   | endothelial            | 0.635     | 4.059          | 0.000 | surface             | 0.30         | True             |
| 5   | 5.0  | LUM     | stellate_myofibroblast | 0.624     | 2.771          | 0.000 | secreted            | 0.10         | True             |
| 6   | 6.0  | DCN     | stellate_myofibroblast | 0.612     | 2.094          | 0.000 | secreted            | 0.10         | True             |
| 7   | 7.0  | TREM2   | macrophage_monocyte    | 0.606     | 0.694          | 0.028 | surface             | 0.60         | True             |
| 8   | 8.0  | PLVAP   | endothelial            | 0.581     | 0.239          | 0.376 | surface             | 0.40         | True             |
| 11  | 9.0  | COL3A1  | stellate_myofibroblast | 0.553     | 3.555          | 0.000 | secreted            | 0.20         | True             |
| 13  | 10.0 | MMP2    | stellate_myofibroblast | 0.542     | 3.328          | 0.000 | secreted            | 0.55         | True             |
| 14  | 11.0 | C1S     | stellate_myofibroblast | 0.540     | 2.253          | 0.000 | secreted            | 0.10         | True             |
| 15  | 12.0 | ITGBL1  | stellate_myofibroblast | 0.540     | 0.577          | 0.220 | secreted            | 0.10         | True             |
| 16  | 13.0 | EFEMP1  | stellate_myofibroblast | 0.540     | 3.329          | 0.000 | secreted            | 0.10         | True             |
| 22  | 14.0 | COL15A1 | endothelial            | 0.522     | 0.634          | 0.135 | secreted            | 0.10         | True             |
| 25  | 15.0 | RGCC    | endothelial            | 0.512     | 0.351          | 0.288 | unknown             | 0.10         | True             |
| 26  | 16.0 | SPARCL1 | endothelial            | 0.511     | 1.901          | 0.000 | secreted            | 0.10         | True             |
| 63  | 17.0 | PLXDC2  | macrophage_monocyte    | 0.462     | 0.315          | 0.106 | surface             | 0.10         | True             |
| 97  | 18.0 | MGAT4A  | macrophage_monocyte    | 0.430     | 0.579          | 0.008 | secreted            | 0.10         | True             |
| 99  | 19.0 | RGS10   | macrophage_monocyte    | 0.428     | 0.404          | 0.028 | unknown             | 0.10         | True             |
| 141 | 20.0 | IER3    | macrophage_monocyte    | 0.406     | 0.939          | 0.000 | unknown             | 0.10         | True             |

## Known-positive recall (sanity check)

Does the pipeline recover biomarkers already established in the human-liver-fibrosis literature, and where a known positive is missed, *why* (non-significant / filtered / absent)? Panel: `resources/known_positive_markers.tsv`.

|   | metric                          | value  |
|---|---------------------------------|--------|
| 0 | n_known_positives               | 37.000 |
| 1 | recall_selected                 | 8.000  |
| 2 | recall_selected_frac            | 0.216  |
| 3 | surfaced_as_candidate           | 25.000 |
| 4 | detected_significant_in_primary | 26.000 |
| 5 | rescued_by_validation_only      | 1.000  |

|   | gene   | compartment            | found_status             | rank | primary_padj | validation_status | rescued_by_validation |
|---|--------|------------------------|--------------------------|------|--------------|-------------------|-----------------------|
| 0 | COL1A1 | stellate_myofibroblast | scored_not_selected      | NaN  | 3.664619e-31 | val_tested_nonsig | False                 |
| 1 | COL1A2 | stellate_myofibroblast | scored_not_selected      | NaN  | 1.289774e-12 | val_tested_nonsig | False                 |
| 2 | COL3A1 | stellate_myofibroblast | selected                 | 9.0  | 9.999196e-16 | val_tested_nonsig | False                 |
| 3 | COL6A3 | stellate_myofibroblast | DE_tested_nonsignificant | NaN  | 2.988873e-01 | val_tested_nonsig | False                 |
| 4 | COL5A1 | stellate_myofibroblast | scored_not_selected      | NaN  | 3.243491e-08 | val_tested_nonsig | False                 |
| 5 | ACTA2  | stellate_myofibroblast | DE_tested_nonsignificant | NaN  | 4.781326e-01 | val_tested_nonsig | False                 |
| 6 | TAGLN  | stellate_myofibroblast | DE_tested_nonsignificant | NaN  | 5.233530e-01 | val_tested_nonsig | False                 |

|    | gene   | compartment            | found_status             | rank | primary_padj | validation_status | rescued_by_validation |
|----|--------|------------------------|--------------------------|------|--------------|-------------------|-----------------------|
| 7  | TIMP1  | stellate_myofibroblast | scored_not_selected      | NaN  | 1.927909e-11 | val_tested_nonsig | False                 |
| 8  | PDGFRB | stellate_myofibroblast | DE_tested_nonsignificant | NaN  | 9.113876e-01 | val_tested_nonsig | False                 |
| 9  | PDGFRA | stellate_myofibroblast | selected                 | 1.0  | 7.830948e-10 | val_tested_nonsig | False                 |
| 10 | PDGFD  | stellate_myofibroblast | scored_not_selected      | NaN  | 2.669875e-03 | validated         | False                 |
| 11 | LOXL1  | stellate_myofibroblast | scored_not_selected      | NaN  | 1.476092e-03 | val_tested_nonsig | False                 |
| 12 | LOXL2  | stellate_myofibroblast | scored_not_selected      | NaN  | 7.423372e-03 | val_tested_nonsig | False                 |
| 13 | TGFB1  | stellate_myofibroblast | scored_not_selected      | NaN  | 5.980219e-03 | val_tested_nonsig | False                 |
| 14 | VCAN   | stellate_myofibroblast | scored_not_selected      | NaN  | 1.477132e-12 | val_tested_nonsig | False                 |
| 15 | DCN    | stellate_myofibroblast | selected                 | 6.0  | 4.732443e-06 | val_tested_nonsig | False                 |
| 16 | LUM    | stellate_myofibroblast | selected                 | 5.0  | 1.927909e-11 | val_tested_nonsig | False                 |
| 17 | SPARC  | stellate_myofibroblast | scored_not_selected      | NaN  | 3.742002e-02 | val_tested_nonsig | False                 |
| 18 | CCN2   | stellate_myofibroblast | absent_primary           | NaN  | NaN          | val_absent        | False                 |
| 19 | FN1    | stellate_myofibroblast | scored_not_selected      | NaN  | 1.849887e-03 | val_tested_nonsig | False                 |
| 20 | SMOC2  | stellate_myofibroblast | DE_tested_nonsignificant | NaN  | 8.753296e-01 | validated         | True                  |
| 21 | THY1   | stellate_myofibroblast | scored_not_selected      | NaN  | 9.999196e-16 | val_tested_nonsig | False                 |
| 22 | MMP2   | stellate_myofibroblast | selected                 | 10.0 | 2.200474e-09 | val_tested_nonsig | False                 |
| 23 | TREM2  | macrophage_monocyte    | selected                 | 7.0  | 2.765572e-02 | val_absent        | False                 |
| 24 | CD9    | macrophage_monocyte    | scored_not_selected      | NaN  | 2.941762e-03 | val_tested_nonsig | False                 |
| 25 | SPP1   | macrophage_monocyte    | scored_not_selected      | NaN  | 1.362951e-01 | val_tested_nonsig | False                 |
| 26 | GPNMB  | macrophage_monocyte    | DE_tested_nonsignificant | NaN  | 8.999708e-01 | val_tested_nonsig | False                 |
| 27 | FABP5  | macrophage_monocyte    | DE_tested_nonsignificant | NaN  | 8.858496e-01 | val_tested_nonsig | False                 |
| 28 | CD63   | macrophage_monocyte    | DE_tested_nonsignificant | NaN  | 9.924401e-01 | val_tested_nonsig | False                 |
| 29 | LGMN   | macrophage_monocyte    | DE_significant_wrong_dir | NaN  | 2.087164e-02 | val_tested_nonsig | False                 |
| 30 | PLVAP  | endothelial            | selected                 | 8.0  | 3.761008e-01 | val_tested_nonsig | False                 |
| 31 | ACKR1  | endothelial            | selected                 | 4.0  | 1.200014e-07 | val_absent        | False                 |
| 32 | VWA1   | endothelial            | scored_not_selected      | NaN  | 1.471575e-01 | val_tested_nonsig | False                 |
| 33 | CD34   | endothelial            | DE_tested_nonsignificant | NaN  | 9.528367e-01 | val_tested_nonsig | False                 |
| 34 | KRT19  | cholangiocyte          | scored_not_selected      | NaN  | 9.376427e-01 | val_tested_nonsig | False                 |
| 35 | KRT7   | cholangiocyte          | scored_not_selected      | NaN  | 4.204185e-01 | val_tested_nonsig | False                 |
| 36 | SOX9   | cholangiocyte          | DE_tested_nonsignificant | NaN  | 9.971689e-01 | val_tested_nonsig | False                 |

## Niche (subpopulation) structure

Disease-associated subclusters within each compartment (donor-aware enrichment, valid within a sorted compartment); their one-vs-rest markers are the subset biomarkers folded into the ranked list above, recovering signals (PLVAP, TREM2, ...) that compartment-level pseudobulk buries.

### gse136103

|    | compartment            | subcluster               | n_cells | frac_test | frac_ref | disease_log2fc | disease_associated |
|----|------------------------|--------------------------|---------|-----------|----------|----------------|--------------------|
| 9  | stellate_myofibroblast | stellate_myofibroblast:8 | 50      | 0.1089    | 0.0035   | 4.613          | True               |
| 44 | lymphoid               | lymphoid:10              | 95      | 0.0072    | 0.0003   | 2.710          | True               |
| 23 | endothelial            | endothelial:10           | 236     | 0.0627    | 0.0124   | 2.249          | True               |
| 32 | endothelial            | endothelial:9            | 291     | 0.0539    | 0.0116   | 2.119          | True               |
| 27 | endothelial            | endothelial:4            | 647     | 0.1040    | 0.0269   | 1.911          | True               |
| 21 | endothelial            | endothelial:0            | 1093    | 0.2446    | 0.0660   | 1.874          | True               |
| 22 | endothelial            | endothelial:1            | 975     | 0.2054    | 0.0764   | 1.416          | True               |
| 16 | macrophage_monocyte    | macrophage_monocyte:5    | 550     | 0.1167    | 0.0459   | 1.328          | True               |
| 1  | stellate_myofibroblast | stellate_myofibroblast:1 | 258     | 0.5364    | 0.2283   | 1.229          | True               |
| 50 | lymphoid               | lymphoid:7               | 1130    | 0.0575    | 0.0258   | 1.124          | True               |
| 37 | cholangiocyte          | cholangiocyte:4          | 191     | 0.1687    | 0.0799   | 1.069          | True               |
| 31 | endothelial            | endothelial:8            | 305     | 0.0505    | 0.0255   | 0.958          | True               |
| 8  | stellate_myofibroblast | stellate_myofibroblast:7 | 95      | 0.1592    | 0.0892   | 0.829          | True               |
| 15 | macrophage_monocyte    | macrophage_monocyte:4    | 753     | 0.1478    | 0.0835   | 0.816          | True               |
| 34 | cholangiocyte          | cholangiocyte:1          | 246     | 0.2533    | 0.1565   | 0.691          | True               |
| 46 | lymphoid               | lymphoid:3               | 4756    | 0.2080    | 0.1343   | 0.628          | True               |

### gse244832

|    | compartment            | subcluster               | n_cells | frac_test | frac_ref | disease_log2fc | disease_associated |
|----|------------------------|--------------------------|---------|-----------|----------|----------------|--------------------|
| 8  | stellate_myofibroblast | stellate_myofibroblast:8 | 220     | 0.4255    | 0.0000   | 8.737          | True               |
| 38 | hepatocyte             | hepatocyte:8             | 219     | 0.0711    | 0.0007   | 5.372          | True               |
| 20 | endothelial            | endothelial:10           | 266     | 0.0685    | 0.0037   | 3.890          | True               |
| 29 | endothelial            | endothelial:9            | 437     | 0.0465    | 0.0048   | 3.023          | True               |
| 36 | hepatocyte             | hepatocyte:6             | 1171    | 0.0331    | 0.0044   | 2.660          | True               |
| 3  | stellate_myofibroblast | stellate_myofibroblast:3 | 1085    | 0.1218    | 0.0507   | 1.249          | True               |
| 4  | stellate_myofibroblast | stellate_myofibroblast:4 | 943     | 0.1233    | 0.0535   | 1.189          | True               |
| 16 | macrophage_monocyte    | macrophage_monocyte:6    | 135     | 0.0544    | 0.0246   | 1.111          | True               |
| 28 | endothelial            | endothelial:8            | 470     | 0.0348    | 0.0209   | 0.707          | True               |

## Cross-dataset reproducibility

|    | gene     | compartment            | primary_lfc | primary_padj | valid_lfc | valid_padj | repro_score |
|----|----------|------------------------|-------------|--------------|-----------|------------|-------------|
| 0  | COL1A1   | stellate_myofibroblast | 4.201       | 0.0          | 0.002     | 0.997      | 0.6         |
| 1  | COL3A1   | stellate_myofibroblast | 3.555       | 0.0          | 0.018     | 0.929      | 0.6         |
| 2  | RARRES1  | stellate_myofibroblast | 5.189       | 0.0          | 0.015     | 0.925      | 0.6         |
| 3  | THY1     | stellate_myofibroblast | 3.737       | 0.0          | 0.015     | 0.851      | 0.6         |
| 4  | SPON2    | stellate_myofibroblast | 3.678       | 0.0          | 0.018     | 0.949      | 0.6         |
| 5  | MMP23B   | stellate_myofibroblast | 3.902       | 0.0          | 0.019     | 0.885      | 0.6         |
| 6  | SERPINF1 | stellate_myofibroblast | 3.987       | 0.0          | 0.102     | 0.472      | 0.6         |
| 7  | COL1A2   | stellate_myofibroblast | 3.246       | 0.0          | 0.038     | 0.813      | 0.6         |
| 8  | VCAN     | stellate_myofibroblast | 5.709       | 0.0          | 0.182     | 0.320      | 0.6         |
| 9  | LUM      | stellate_myofibroblast | 2.771       | 0.0          | 0.058     | 0.566      | 0.6         |
| 10 | PCOLCE   | stellate_myofibroblast | 3.178       | 0.0          | -0.013    | 0.947      | 0.0         |
| 11 | TIMP1    | stellate_myofibroblast | 3.680       | 0.0          | 0.034     | 0.750      | 0.6         |
| 12 | IGF1     | stellate_myofibroblast | 5.306       | 0.0          | 0.015     | 0.873      | 0.6         |
| 13 | F2R      | stellate_myofibroblast | 3.704       | 0.0          | 0.072     | 0.550      | 0.6         |
| 14 | PTGIS    | stellate_myofibroblast | 4.439       | 0.0          | 0.017     | 0.918      | 0.6         |
| 15 | PDGFRA   | stellate_myofibroblast | 3.608       | 0.0          | 0.653     | 0.055      | 1.0         |
| 16 | FBLN1    | stellate_myofibroblast | 3.314       | 0.0          | 0.009     | 0.966      | 0.6         |
| 17 | MEG3     | stellate_myofibroblast | 2.530       | 0.0          | -0.015    | 0.951      | 0.0         |
| 18 | EFEMP1   | stellate_myofibroblast | 3.329       | 0.0          | 1.744     | 0.005      | 1.0         |
| 19 | MMP2     | stellate_myofibroblast | 3.328       | 0.0          | 0.037     | 0.740      | 0.6         |

## Mechanism: pathways and cell-cell communication

### gse136103

|    | compartment            | condition | source   | mean_activity | level           |
|----|------------------------|-----------|----------|---------------|-----------------|
| 0  | cholangiocyte          | cirrhotic | Androgen | 1.0598        | pathway_progeny |
| 1  | cholangiocyte          | healthy   | Androgen | 0.8873        | pathway_progeny |
| 2  | endothelial            | cirrhotic | Androgen | 0.4854        | pathway_progeny |
| 3  | endothelial            | healthy   | Androgen | 1.2176        | pathway_progeny |
| 4  | lymphoid               | cirrhotic | Androgen | 0.3709        | pathway_progeny |
| 5  | lymphoid               | healthy   | Androgen | 0.3829        | pathway_progeny |
| 6  | macrophage_monocyte    | cirrhotic | Androgen | 0.5556        | pathway_progeny |
| 7  | macrophage_monocyte    | healthy   | Androgen | 0.6782        | pathway_progeny |
| 8  | myeloid_other          | cirrhotic | Androgen | 0.6113        | pathway_progeny |
| 9  | myeloid_other          | healthy   | Androgen | 0.6975        | pathway_progeny |
| 10 | stellate_myofibroblast | cirrhotic | Androgen | 0.2392        | pathway_progeny |
| 11 | stellate_myofibroblast | healthy   | Androgen | 0.0296        | pathway_progeny |
| 12 | cholangiocyte          | cirrhotic | EGFR     | 2.2357        | pathway_progeny |
| 13 | cholangiocyte          | healthy   | EGFR     | 1.9298        | pathway_progeny |
| 14 | endothelial            | cirrhotic | EGFR     | 2.4658        | pathway_progeny |

|   | source        | target        | ligand_complex | receptor_complex | fibrotic_condition | delta_specificity | enriched_in_fibrosis |
|---|---------------|---------------|----------------|------------------|--------------------|-------------------|----------------------|
| 0 | myeloid_other | cholangiocyte | SCT            | SCTR             | cirrhotic          | 5.060615          | True                 |
| 1 | cholangiocyte | lymphoid      | DEFB1          | CCR6             | cirrhotic          | 4.847387          | True                 |
| 2 | cholangiocyte | endothelial   | CXCL6          | ACKR1            | cirrhotic          | 4.320696          | True                 |

|    | source                 | target                 | ligand_complex | receptor_complex | fibrotic_condition | delta_specificity | enriched_in_fibrosis |
|----|------------------------|------------------------|----------------|------------------|--------------------|-------------------|----------------------|
| 3  | stellate_myofibroblast | endothelial            | COL1A1         | ITGA10_ITGB1     | cirrhotic          | 4.167834          | True                 |
| 4  | stellate_myofibroblast | endothelial            | COL1A1         | FLT4             | cirrhotic          | 4.097689          | True                 |
| 5  | stellate_myofibroblast | endothelial            | COL1A1         | ITGA5            | cirrhotic          | 3.917435          | True                 |
| 6  | stellate_myofibroblast | endothelial            | COL3A1         | ITGA10_ITGB1     | cirrhotic          | 3.907577          | True                 |
| 7  | stellate_myofibroblast | endothelial            | COL1A1         | CD93             | cirrhotic          | 3.897793          | True                 |
| 8  | stellate_myofibroblast | cholangiocyte          | COL1A1         | DDR1             | cirrhotic          | 3.707833          | True                 |
| 9  | stellate_myofibroblast | stellate_myofibroblast | COL1A1         | ITGA1_ITGB1      | cirrhotic          | 3.691121          | True                 |
| 10 | stellate_myofibroblast | stellate_myofibroblast | COL1A1         | DDR2             | cirrhotic          | 3.682846          | True                 |
| 11 | stellate_myofibroblast | cholangiocyte          | COL1A1         | ITGAV_ITGB8      | cirrhotic          | 3.674624          | True                 |
| 12 | stellate_myofibroblast | stellate_myofibroblast | COL1A2         | ITGA1_ITGB1      | cirrhotic          | 3.674624          | True                 |
| 13 | stellate_myofibroblast | stellate_myofibroblast | THBS2          | NOTCH3           | cirrhotic          | 3.581461          | True                 |
| 14 | endothelial            | endothelial            | CCL14          | ACKR1            | cirrhotic          | 3.581461          | True                 |

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|    | compartment            | condition | source   | mean_activity | level           |
|----|------------------------|-----------|----------|---------------|-----------------|
| 0  | cholangiocyte          | MASH      | Androgen | 3.1403        | pathway_progeny |
| 1  | cholangiocyte          | MASL      | Androgen | 2.6690        | pathway_progeny |
| 2  | cholangiocyte          | normal    | Androgen | 1.8914        | pathway_progeny |
| 3  | endothelial            | MASH      | Androgen | 2.1593        | pathway_progeny |
| 4  | endothelial            | MASL      | Androgen | 2.2036        | pathway_progeny |
| 5  | endothelial            | normal    | Androgen | 2.4728        | pathway_progeny |
| 6  | hepatocyte             | MASH      | Androgen | 4.8672        | pathway_progeny |
| 7  | hepatocyte             | MASL      | Androgen | 4.7303        | pathway_progeny |
| 8  | hepatocyte             | normal    | Androgen | 4.5555        | pathway_progeny |
| 9  | macrophage_monocyte    | MASH      | Androgen | 2.2043        | pathway_progeny |
| 10 | macrophage_monocyte    | MASL      | Androgen | 1.8532        | pathway_progeny |
| 11 | macrophage_monocyte    | normal    | Androgen | 1.6571        | pathway_progeny |
| 12 | stellate_myofibroblast | MASH      | Androgen | 2.5075        | pathway_progeny |
| 13 | stellate_myofibroblast | MASL      | Androgen | 2.4753        | pathway_progeny |
| 14 | stellate_myofibroblast | normal    | Androgen | 2.4165        | pathway_progeny |

|    | source                 | target                 | ligand_complex | receptor_complex | fibrotic_condition | delta_specificity | enriched_in_fibrosis |
|----|------------------------|------------------------|----------------|------------------|--------------------|-------------------|----------------------|
| 0  | cholangiocyte          | cholangiocyte          | EFNA5          | EPHA6            | MASH               | 4.776138          | True                 |
| 1  | stellate_myofibroblast | cholangiocyte          | LAMA2          | ITGA2_ITGB1      | MASH               | 4.001292          | True                 |
| 2  | endothelial            | cholangiocyte          | FGF23          | FGFR2            | MASH               | 3.972475          | True                 |
| 3  | stellate_myofibroblast | cholangiocyte          | FGF7           | FGFR2            | MASH               | 3.863243          | True                 |
| 4  | stellate_myofibroblast | cholangiocyte          | LAMA2          | ITGA6_ITGB4      | MASH               | 3.750497          | True                 |
| 5  | cholangiocyte          | stellate_myofibroblast | EFNA5          | EPHA3            | MASH               | 3.421758          | True                 |
| 6  | endothelial            | cholangiocyte          | EFNB2          | EPHA6            | MASH               | 3.281674          | True                 |
| 7  | cholangiocyte          | cholangiocyte          | EFNA5          | EPHB6            | MASH               | 3.266454          | True                 |
| 8  | cholangiocyte          | cholangiocyte          | LAMC2          | ITGAV_ITGB8      | MASH               | 3.191933          | True                 |
| 9  | stellate_myofibroblast | cholangiocyte          | COL3A1         | ITGA2_ITGB1      | MASH               | 3.082785          | True                 |
| 10 | stellate_myofibroblast | cholangiocyte          | LAMC3          | ITGA2_ITGB1      | MASH               | 2.967011          | True                 |
| 11 | endothelial            | cholangiocyte          | HSPG2          | ITGA2            | MASH               | 2.922234          | True                 |
| 12 | stellate_myofibroblast | cholangiocyte          | COL6A3         | ITGA2_ITGB1      | MASH               | 2.785301          | True                 |
| 13 | stellate_myofibroblast | cholangiocyte          | COL4A1         | ITGA2_ITGB1      | MASH               | 2.724684          | True                 |
| 14 | cholangiocyte          | stellate_myofibroblast | NCAM1          | CACNA1C          | MASH               | 2.711973          | True                 |